JEPPIAAR INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Self-Belief | Self Discipline | Self Respect

Kunnam, Sunguvarchatram, Sriperumbudur-631604

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

ACS302 – CLOUD COMPUTING LABORATORY

NAME :

REG NO. :

YEAR : I

SEMESTER : 02

BRANCH :

BONAFIDE CERTIFICATE

This is a certified Bonafide Record Work of Mr./Ms._____

Register No._____submitted for the Anna University Practical Examination held on_____in ACS302 **CLOUD COMPUTING LABORATORY** during the year 2024-2025.

Signature of the Lab In-charge

Head of the Department

Internal Examiner

External Examiner

INSTITUTE VISION:

Jeppiaar Institute of Technology aspires to provide technical education in futuristic technologies with the perspective of innovative, industrial and social application for the betterment of humanity.

INSTITUTE MISSION:

- **M1:** To produce competent and disciplined high-quality professionals with the practical skills necessary to excel as innovative professionals and entrepreneurs for the benefit of the society.
- **M2:** To improve the quality of education through excellence in teaching and learning, research, leadership and by promoting the principles of scientific analysis, and creative thinking.
- **M3:** To provide excellent infrastructure, serene and stimulating environment that is most conducive to learning.
- **M4:** To strive for productive partnership between the Industry and the Institute for research and development in the emerging fields and creating opportunities for employability.
- **M5:** To serve the global community by instilling ethics, values and life skills among the students needed to enrich their lives.

VISION

- The department will serve as a centre of excellence in practicing, training and implementing AI and AI associated techniques that will enable /support innovative thoughts and ideas across industries and society

MISSION

- M1: To collaborate with industry and provide the state of the art infrastructural Facilities to meet the global requirements and societal needs for AI.
- M2: Promote learning and development of students in Artificial Intelligence thought leadership, by providing them a suitable infrastructure and Environment, enabling them to grow into successful entrepreneurs.
- M3: To encourage students to pursue higher education and research in the field of AI.
- M4: To impart moral and ethical values in their profession

PROGRAMME EDUCATIONAL OBJECTIVES

- PEO 1: Utilize their proficiencies in the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volumes of data.

- PEO 2: Advance their technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.

- PEO 3: Think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team.

- PEO 4: Design and model AI based solutions to critical problem domains in the real world

- PEO 5: Exhibit innovative thoughts and creative ideas for effective contribution towards economy building

PROGRAM OUTCOMES

Engineering Graduates will be able to:

1. **Engineering knowledge:** (K3) Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:** (K4) Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions:** (K4) Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems:** (K5) Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** (K3, K5, K6) Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society:** (A3) Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:** (A2) Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics:** (A3) Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work:** (A3) Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication:** (A3) Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance:** (A3) Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning:** (A2) Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO 1: To evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.

PSO 2: To arrive at actionable Foresight, Insight, hindsight from data for solving business and engineering problems

PSO 3: To create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems

PSO 4: To develop data analytics and data visualization skills, skills pertaining to knowledge acquisition, knowledge representation and knowledge engineering, and hence be capable of coordinating complex projects.

PSO 5: To able to carry out fundamental research to cater the critical needs of the society through cutting edge technologies of AI.

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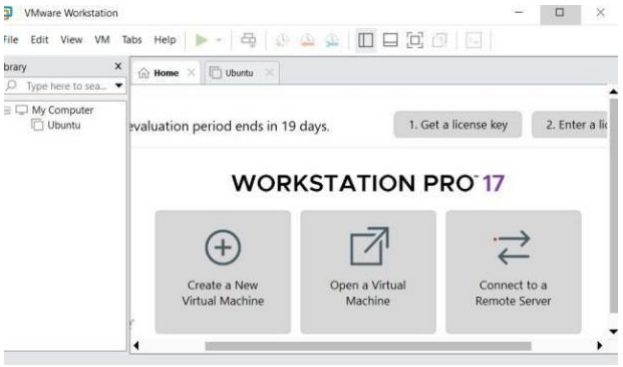
Ex. No:1	Install Guest OS on VMWARE
DATE:	

Aim:

To find the procedure to run VM of different configuration and Allocate memory and storage space as per requirement.

Procedure:

Step 1: Download and Install VMware Workstation Player



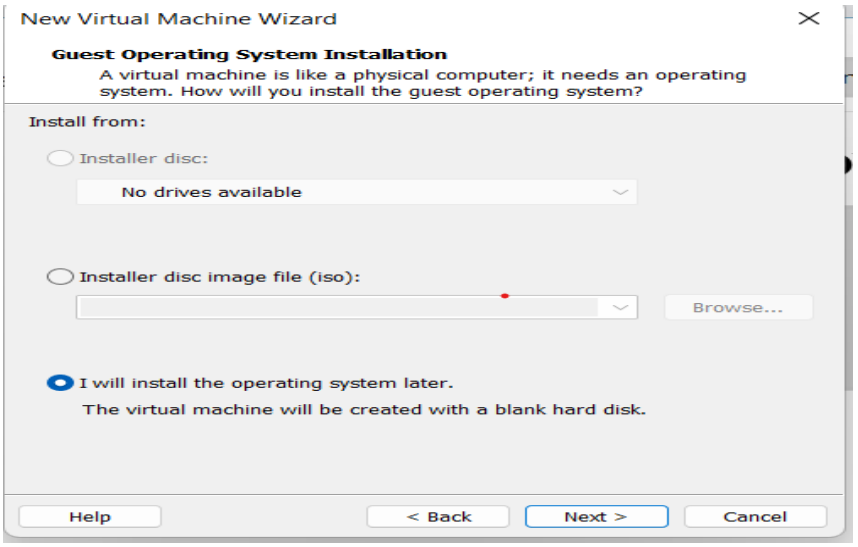
Step 2: Create a New Virtual Machine

Open VMware Workstation Player

Click on "Create a New Virtual Machine" or go to File > New Virtual Machine

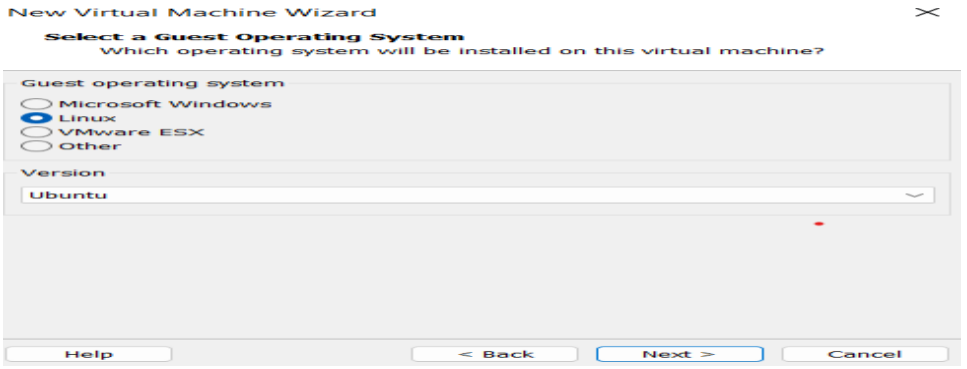
Step 3: Virtual Machine Configuration Wizard

The Virtual machine configuration Wizard will appear. Choose “Typical” configuration and click “Next”.



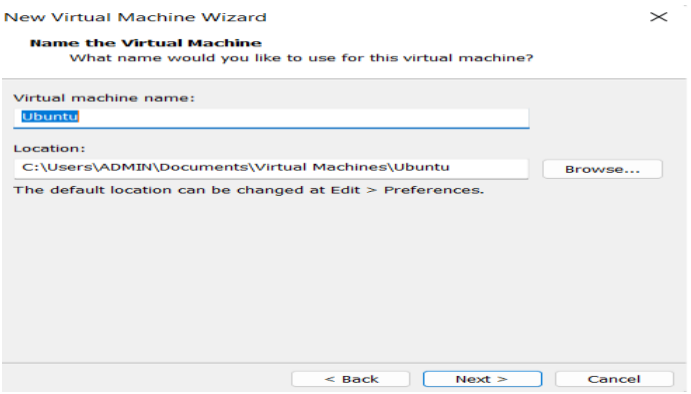
Step 4: Install Guest Operating System

- Choose the installation method for the guest OS. You can either install from a disc or image file (ISO) or choose to install later. If you have the ISO file for your guest OS, select it during this step
- Click “Next”



Step 5: Enter Guest OS Details

- Enter the name of your virtual machine and choose the location where you want to save it.
- Select the appropriate guest operating system and version. For example, if you are installing Windows 10, choose "Windows" as the guest OS and "Windows 10 x64" as the version.
- Click "Next."



Step 6: Configure Virtual Machine Hardware

- Allocate memory: Choose how much RAM you want to allocate to the virtual machine. Make sure to leave enough memory for your host OS to run smoothly as well.
- Allocate storage: Choose whether to store the virtual disk as a single file or split into multiple files. Specify the disk size, and you can also choose to allocate all disk space now or let it grow as needed.
- Click "Next."

Step 8: Customize Hardware (Optional)

- If needed, you can customize the virtual machine's hardware settings like CPU cores, network adapters, graphics memory, etc. Otherwise, you can leave them as default.
- Click "Finish" once you are satisfied with the settings.

Step 8: Install Guest OS

- Start the virtual machine you just created. The virtual machine will boot from the ISO or installation media you provided earlier.
- Follow the standard installation process for your guest OS.

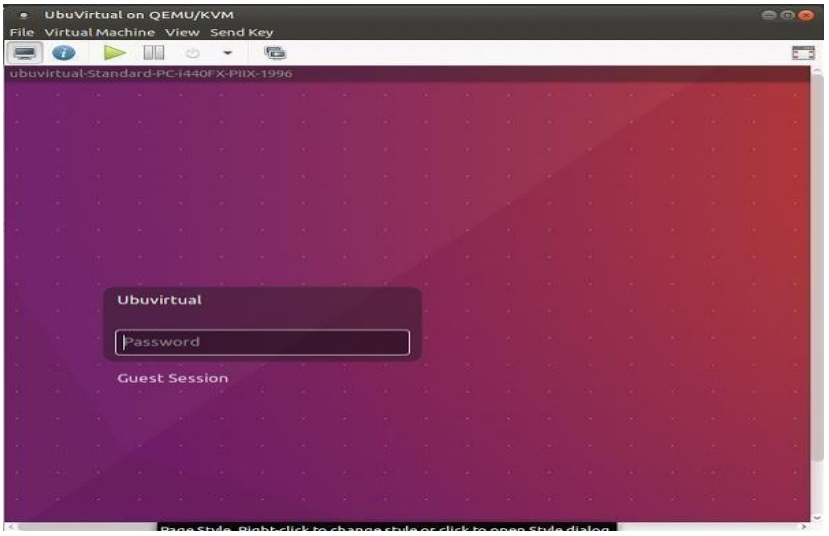
Step 9: Install VMware Tools (Optional but Recommended)

- After installing the guest OS, it is advisable to install VMware Tools within the guest OS. VMware Tools provides better integration between the host and guest OS, enabling features like shared folders, improved graphics, and more.

In the same way, we can install windows OS also. The output screens for the virtual machines of different configurations.

OUTPUT:

(i) Ubuntu Operating System in Virtual Machine



(ii) Windows7 Operating System in Virtual Machine

RESULT:

Thus, various configurations of Virtual machines has been created and run.

Ex. No:2(a)	Shrink and Extend Virtual Disk
DATE:	

Aim:

To find the procedure of Shrink a virtual disk involves reducing its size to reclaim unused space and extend a virtual disk allows you to increase its size to accommodate more data.

Procedure:

Shrink a Virtual Disk:

Step1: Inside the virtual machine, delete unnecessary files and empty the recycle bin/trash to free up space.

Step 2: Defragment the virtual machine's disk to move all the data to the beginning of the disk.

Step 3: Shutdown the virtual machine.

Step 4: On the host system, open the virtualization software

Step 5: Navigate to the virtual machine's settings or configuration.

Step 6: Select the virtual disk you want to shrink.

Step 7: Look for an option to shrink or compact the disk and follow the prompts.

Step 8: The process might take some time, and it's essential to back up your virtual machine before proceeding, as data loss can occur in some cases.

Extend Virtual Disk:

Step 1: Shutdown the virtual machine.

Step 2: On the host system, open the virtualization software (e.g., VMware, VirtualBox).

Step 3: Navigate to the virtual machine's settings or configuration.

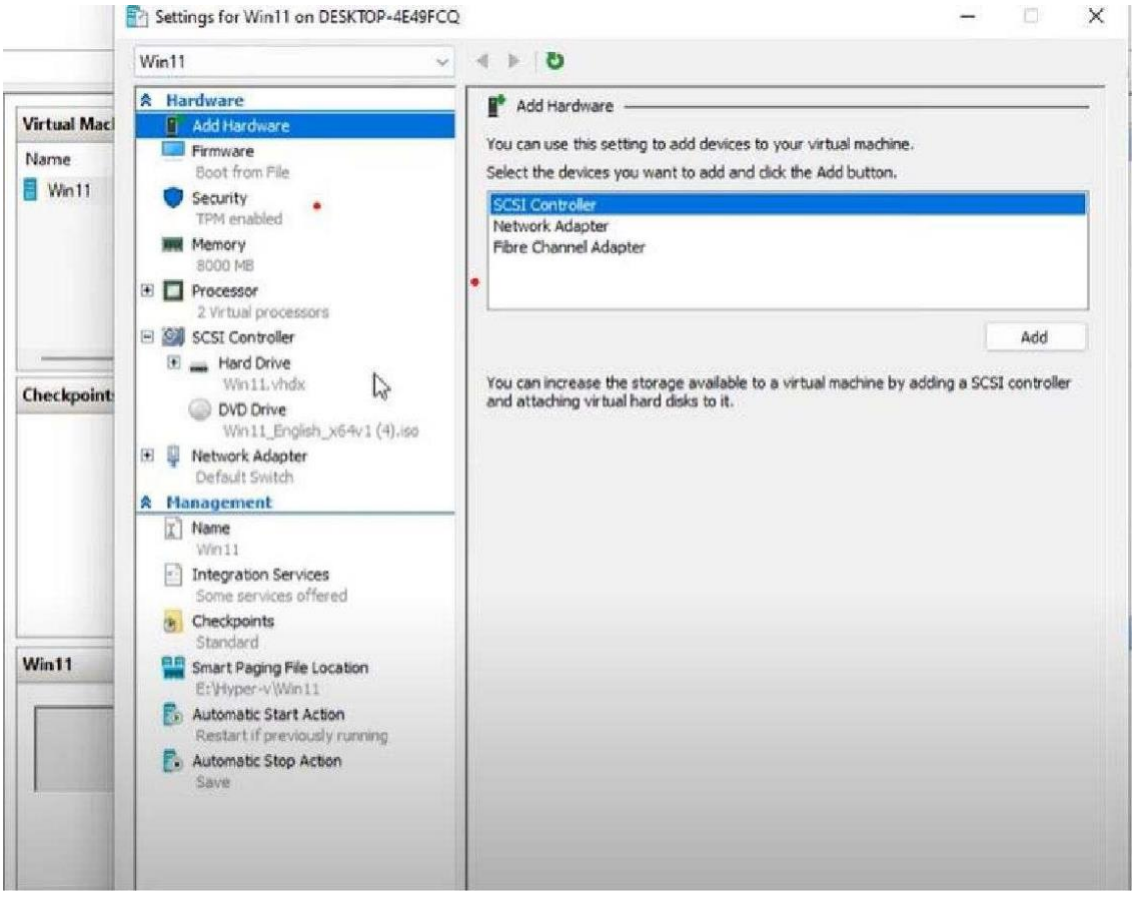
Step 4: Select the virtual disk you want to extend.

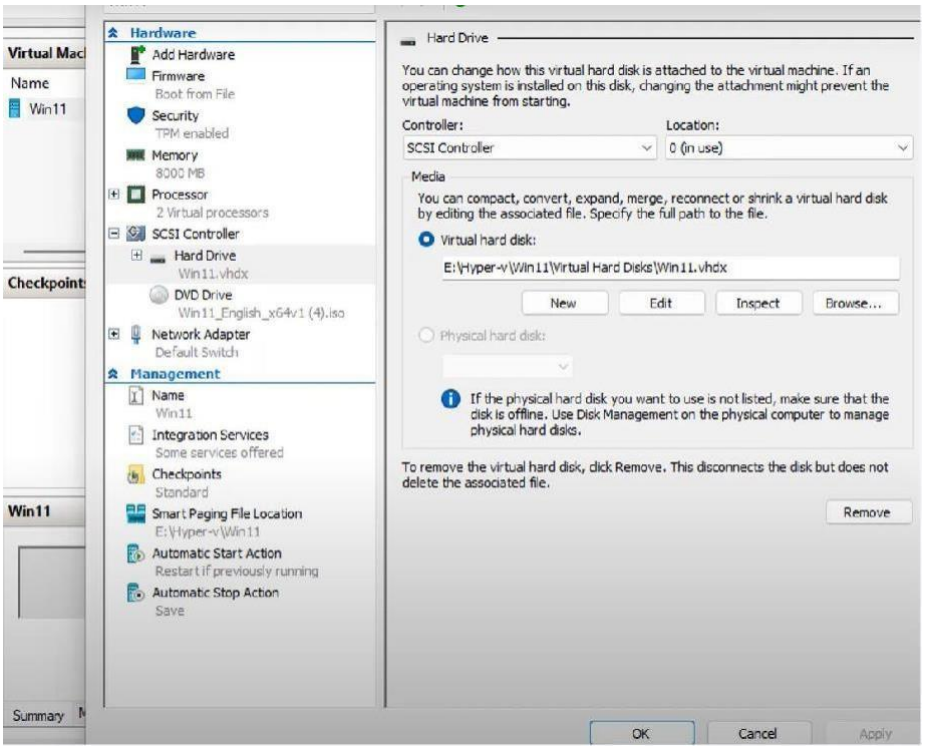
Step 5: Look for an option to increase or extend the disk and follow the prompts.

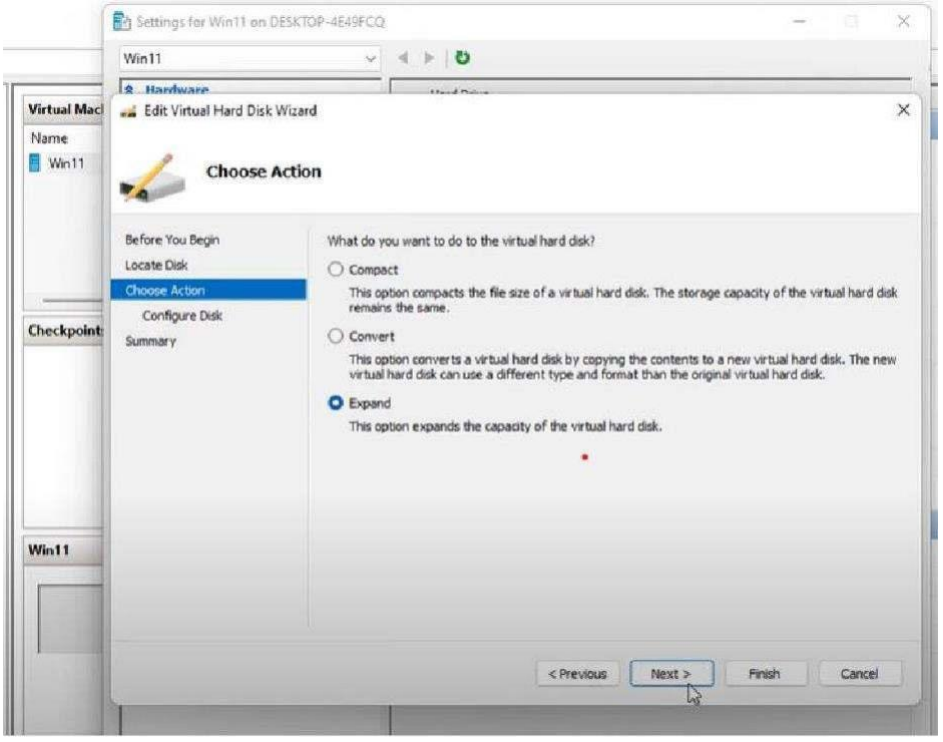
Step 6: Specify the new size for the virtual disk (make sure to allocate enough space for your needs).

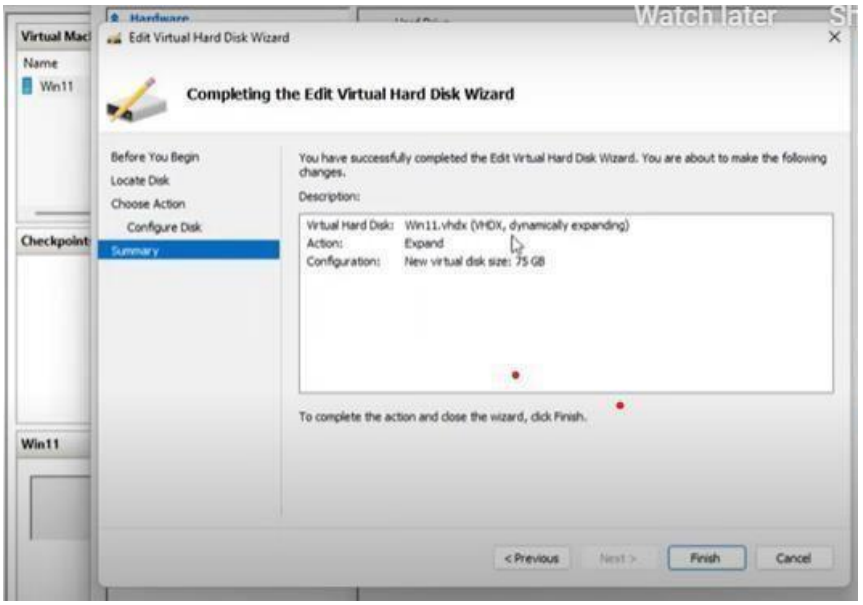
Step 7: The process might take some time, and it's crucial to back up your virtual machine before proceeding, as extending the disk involves modifying the underlying file system.

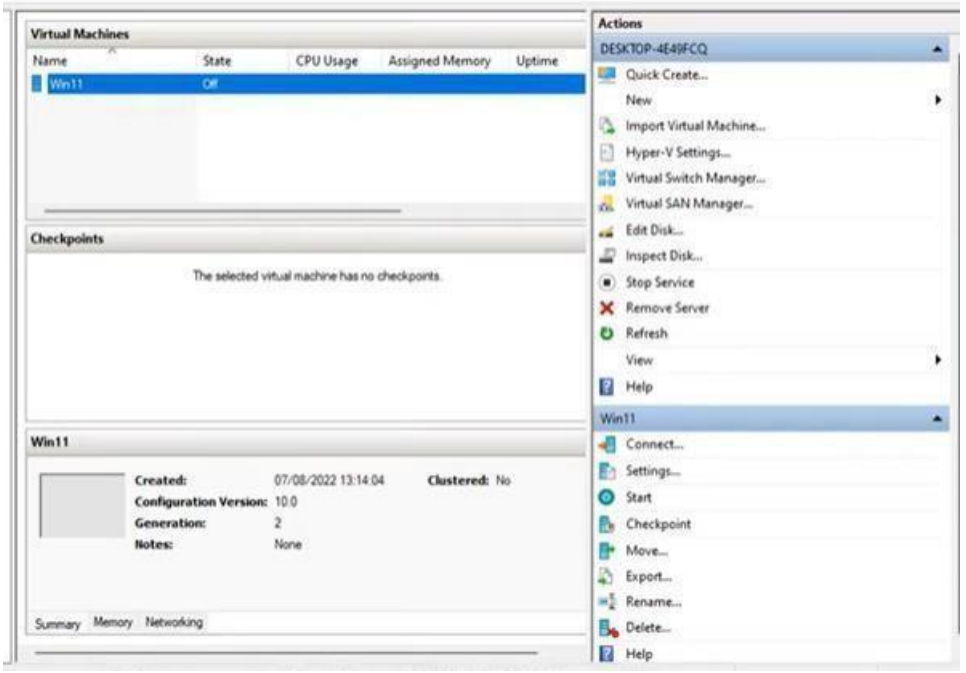
Output:




Hardware
Add Hardware
Firmware
Boot from File
Security
TPM enabled
Memory
8000 MB
Processor
2 Virtual processors
SCSI Controller
Hard Drive
Win11.vhdx
DVD Drive
Win11_English_x64v1 (4).iso
Network Adapter
Default Switch
Management
Name
Win11
Integration Services
Some services offered
Checkpoints
Standard
Smart Paging File Location
E:\Hyper-v\Win11
Automatic Start Action
Restart if previously running
Automatic Stop Action
Save
Hard Drive
You can change how this virtual hard disk is attached to the virtual machine. If an operating system is installed on this disk, changing the attachment might prevent the virtual machine from starting.
Controller:
SCSI Controller
Location:
0 (in use)
Media
You can compact, convert, expand, merge, reconnect or shrink a virtual hard disk by editing the associated file. Specify the full path to the file.
Virtual hard disk:
E:\Hyper-v\Win11\Virtual Hard Disks\Win11.vhdx
New
Edit
Inspect
Browse...
Physical hard disk:
If the physical hard disk you want to use is not listed, make sure that the disk is offline. Use Disk Management on the physical computer to manage physical hard disks.
To remove the virtual hard disk, click Remove. This disconnects the disk but does not delete the associated file.
Remove
OK
Cancel
Apply


Settings for Win11 on DESKTOP-4E49FCQ
Win11
Edit Virtual Hard Disk Wizard
Choose Action
Before You Begin
Locate Disk
Choose Action
Configure Disk
Summary
What do you want to do to the virtual hard disk?
Compact
This option compacts the file size of a virtual hard disk. The storage capacity of the virtual hard disk remains the same.
Convert
This option converts a virtual hard disk by copying the contents to a new virtual hard disk. The new virtual hard disk can use a different type and format than the original virtual hard disk.
Expand
This option expands the capacity of the virtual hard disk.
< Previous
Next >
Finish
Cancel





Result:

Thus, the shrink and extend of virtual disk has been created and run Successfully.

Ex. No:2(b)	Create, Manage, Configure and schedule snapshots
DATE:	

Aim: To find the procedure of Creating, managing, configuring, and scheduling snapshots is a fundamental part of virtual machine management.

Procedure:

Step 1: Creating a Snapshot: To create a snapshot, follow these steps:

a. Select the virtual machine in the management interface (e.g., vSphere Client).

b. Right-click on the virtual machine and choose "Snapshot" or "Take Snapshot."

c. Provide a name and description for the snapshot.

d. Optionally, select the memory state to capture the virtual machine's running state (requires the virtual machine to be powered off).

e. Click "OK" to create the snapshot.

Step 2: Managing Snapshots: Managing snapshots involves tasks such as viewing, reverting, deleting, and consolidating snapshots.

a. Viewing Snapshots: To see the list of snapshots for a virtual machine, navigate to the "Snapshots" tab in the virtual machine's details.

b. Reverting to a Snapshot: To revert a virtual machine to a specific snapshot, right-click on the snapshot and choose "Revert to Snapshot." This action will roll back the virtual machine to the state captured in the selected snapshot.

c. Deleting Snapshots: To delete a snapshot, right-click on the snapshot and choose "Delete Snapshot." You can choose to delete the snapshot only or delete the snapshot and consolidate its changes into the base virtual machine disk.

d. Consolidating Snapshots: If you have multiple snapshots, you can perform a "Snapshot Consolidation" to merge all snapshots into the base disk. This helps to reduce storage requirements and improves performance.

Step 3: Configuring Snapshot Settings: The snapshot settings allow you to define how snapshots are managed and stored.

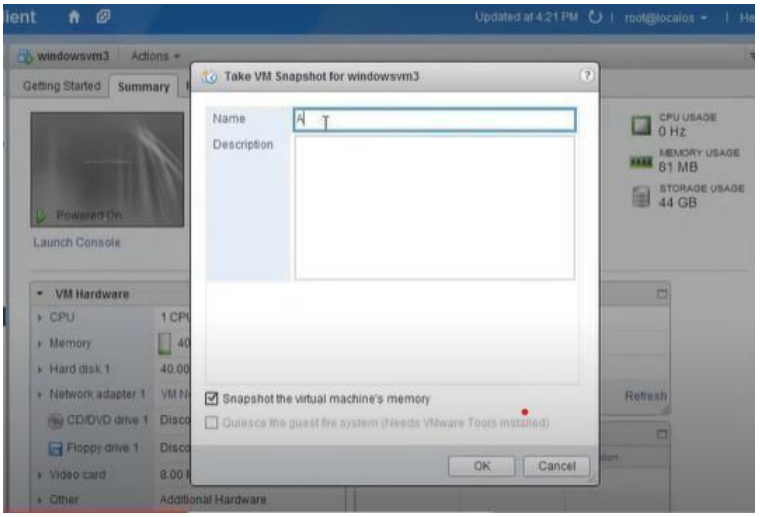
a. Snapshot Location: Configure the location where snapshot files are stored, which can be on the same data store as the virtual machine or on a different storage location.

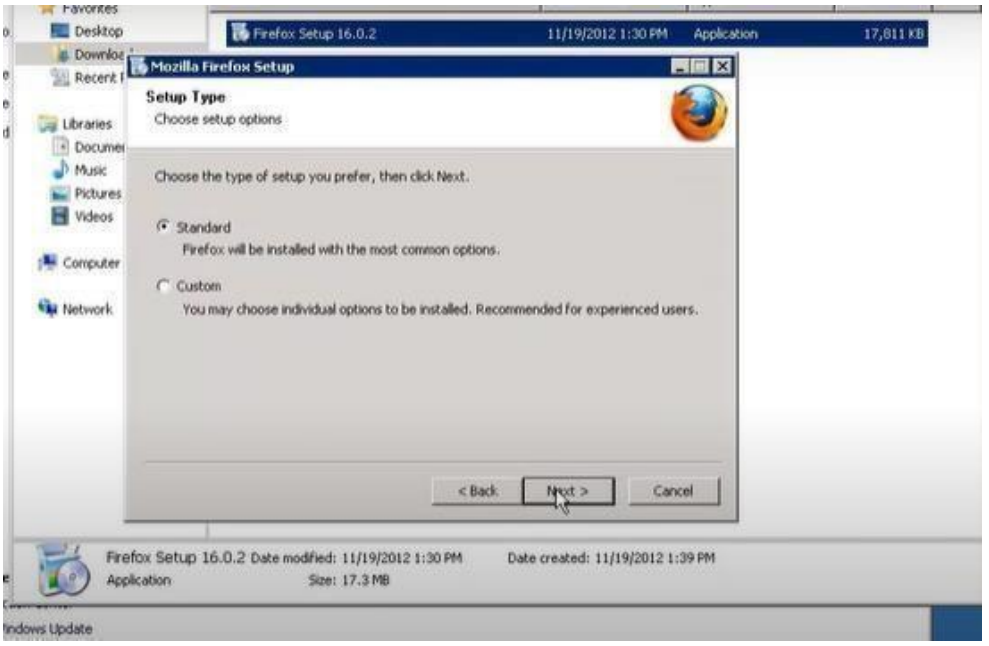
b. Snapshot Quiescent: Enable or disable the use of VMware Tools to quiesce the file system inside the virtual machine before taking a snapshot. This helps ensure application consistency during the snapshot.

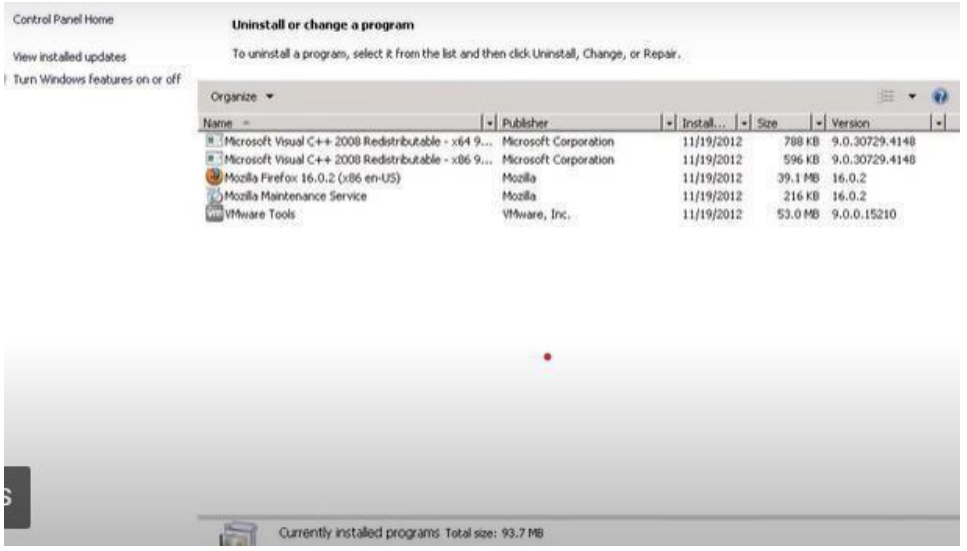
Step 4: Scheduling Snapshots: Some virtualization platforms allow you to schedule snapshots to be taken automatically at specific intervals.

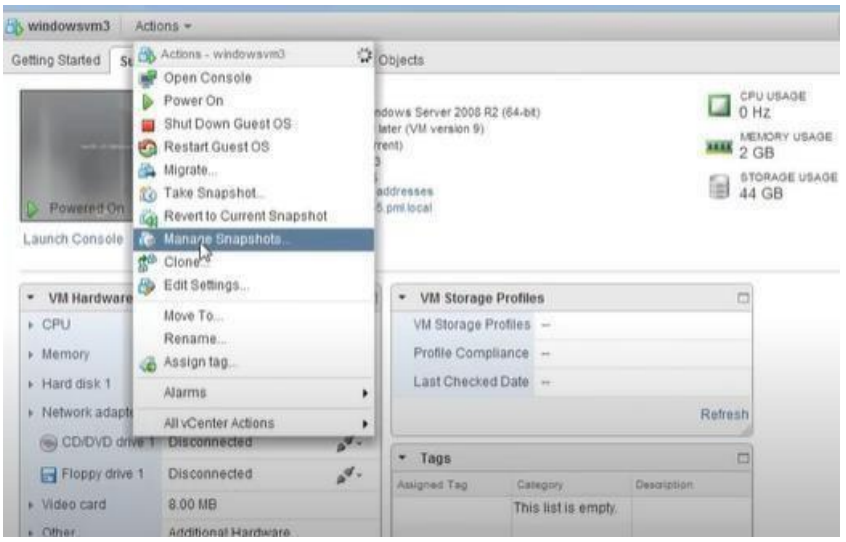
a. Scheduled Snapshots: Navigate to the virtual machine's settings, find the "Snapshot Schedule" option, and configure the frequency and retention policy for automatic snapshots.

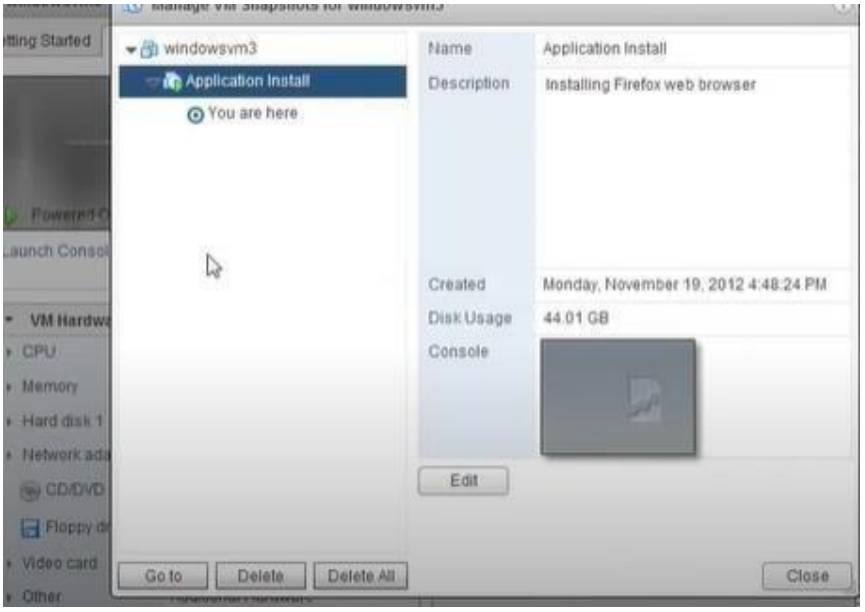
Output:




Firefox Setup 16.0.2
11/19/2012 1:30 PM
Application
17,811 KB
Mozilla Firefox Setup
Setup Type
Choose setup options
Choose the type of setup you prefer, then click Next.
Standard
Firefox will be installed with the most common options.
Custom
You may choose individual options to be installed. Recommended for experienced users.
< Back
Next >
Cancel
Firefox Setup 16.0.2 Date modified: 11/19/2012 1:30 PM
Date created: 11/19/2012 1:39 PM
Application
Size: 17.3 MB


Control Panel Home
View installed updates
Turn Windows features on or off
Uninstall or change a program
To uninstall a program, select it from the list and then click Uninstall, Change, or Repair.
Organize
Name
Publisher
Install...
Size
Version
Microsoft Visual C++ 2008 Redistributable - x64 9...
Microsoft Corporation
11/19/2012
788 KB
9.0.30729.4148
Microsoft Visual C++ 2008 Redistributable - x86 9...
Microsoft Corporation
11/19/2012
596 KB
9.0.30729.4148
Mozilla Firefox 16.0.2 (x86 en-US)
Mozilla
11/19/2012
39.1 MB
16.0.2
Mozilla Maintenance Service
Mozilla
11/19/2012
216 KB
16.0.2
VMware Tools
VMware, Inc.
11/19/2012
53.0 MB
9.0.0.15210
Currently installed programs Total size: 93.7 MB





Result:

Thus, the Create, Manage, Configure and schedule snapshots was Created and Run Successfully.

Ex. No:2(c)	Create Spanned, Mirrored and Striped volume
DATE:	

Aim:

To Create Spanned, Mirrored and Striped volume

Procedure:

Step 1: How to create a spanned volume:

a. Right-click on "This PC" or "My Computer" and select "Manage."

b. In the Computer Management window, click on "Disk Management" under "Storage" in the left pane.

c. Identify the disks you want to use in the spanned volume. They should be unallocated or have empty space available.

d. Right-click on the first disk and choose "New Spanned Volume."

e. Follow the on-screen instructions to select additional disks and allocate space for the spanned volume.

f. Assign a drive letter or mount point to the new volume and format it with a file system.

Step 2: How to create a mirrored volume:

a. Follow steps a to c from the spanned volume creation process to open Disk Management.

b. Right-click on one of the disks you want to use in the mirror and choose "New Mirrored Volume."

c. Select the additional disk(s) to mirror the first disk, and then allocate space for the mirrored volume.

d. Assign a drive letter or mount point to the new volume and format it with a file system. e. Windows will start the synchronization process, where data is copied from the source disk to the mirror disk(s).

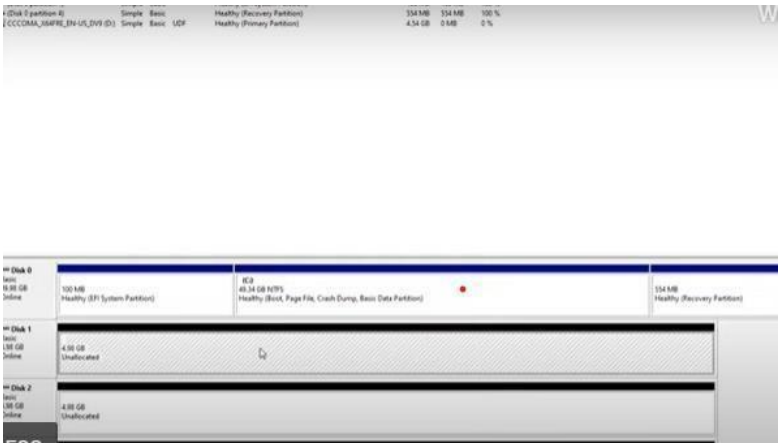
Step 3: How to create a striped volume:

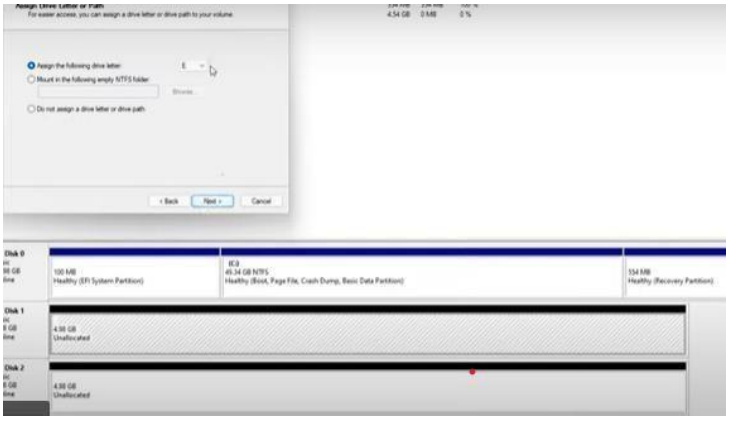
a. Follow steps a to c from the spanned volume creation process to open Disk Management.

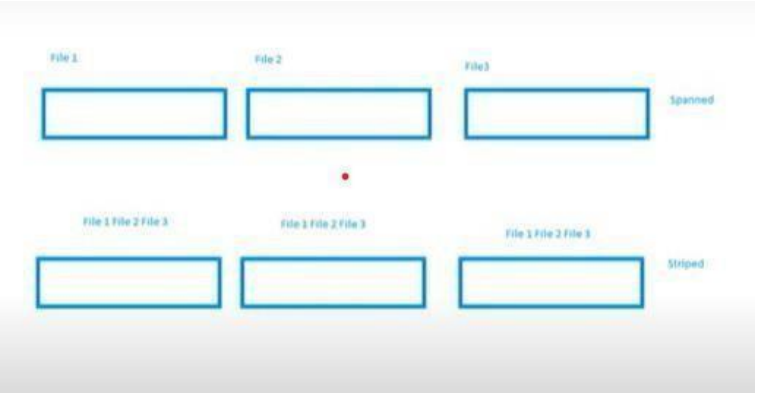
b. Right-click on one of the disks you want to use in the stripe and choose "New Striped Volume."

c. Select the additional disk(s) to include in the stripe, and then allocate space for the striped volume. d. Assign a drive letter or mount point to the new volume and format it with a file system.

Output:







Result:

Thus, The Spanned, Mirrored and Striped volume has been Created and Run Successfully.

Ex. No:2(d)	Create RAID 5 Volume
DATE:	

Aim:

To Create RAID 5 Volume

Procedure:

Step 1: Backup Data: Before creating the RAID 5 volume, back up all your critical data to ensure it is safe in case of any unforeseen issues.

Step 2: Initialize and Convert Disks to Dynamic: If your disks are currently basic disks, you need to convert them to dynamic disks to create a RAID 5 volume. Follow these steps:

a. Right-click on "This PC" or "My Computer" and select "Manage."

b. In the Computer Management window, click on "Disk Management" under "Storage" in the left pane.

c. Identify the disks you want to use in the RAID 5 array. They should be unallocated or have empty space available.

d. Right-click on each disk and select "Convert to Dynamic Disk." Repeat this for all the disks you want to include in the RAID 5 array.

Step 3: Create the RAID 5 Volume: Once you have converted the disks to dynamic, you can proceed to create the RAID 5 volume:

a. Right-click on one of the disks you want to use in the RAID 5 array.

b. Select "New Spanned Volume."

c. Follow the on-screen instructions to select the other disks you want to include in the RAID 5 array. There should be at least three disks in total.

d. Allocate the desired amount of space for the RAID 5 volume. RAID 5 requires a minimum of three disks, and the volume size will be the total capacity of all disks minus one disk's worth of space. For example, if you have three 1TB disks, the RAID 5 volume will have a total capacity of 2TB.

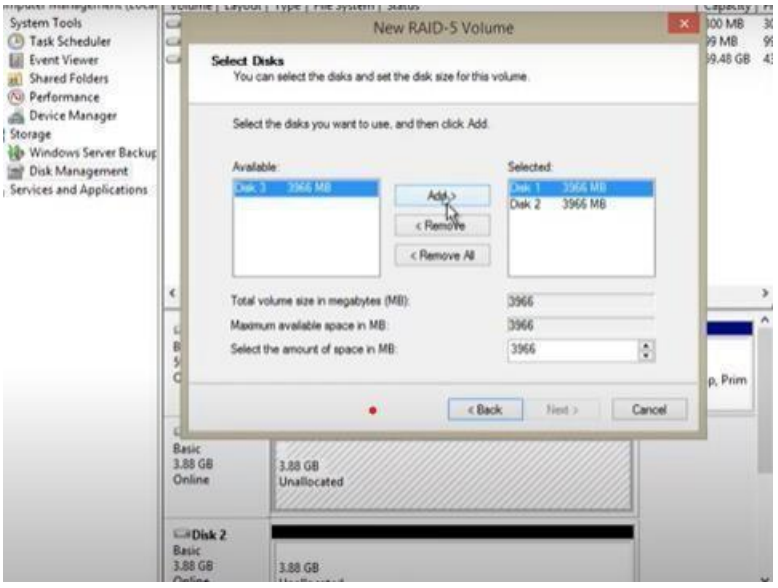
e. Assign a drive letter or mount point to the new RAID 5 volume and format it with a file system.

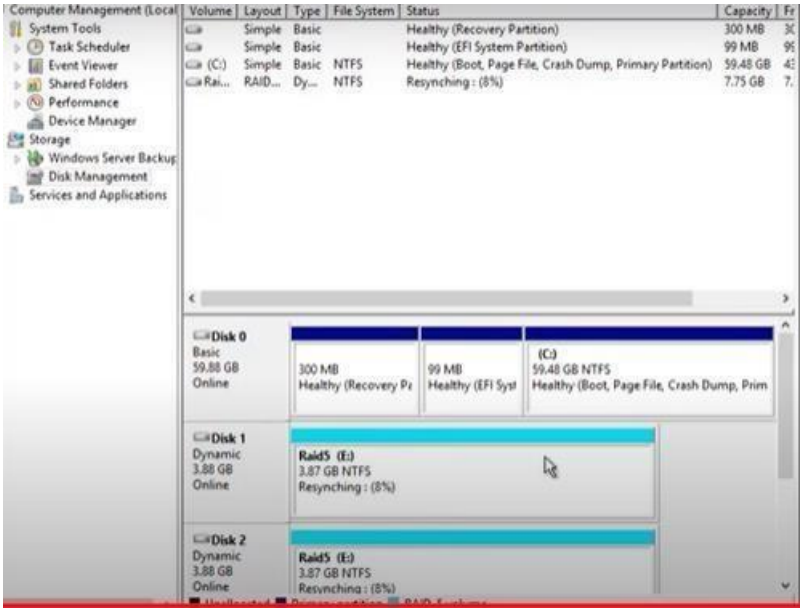
Step 4: Initialize and Format the RAID 5 Volume: After creating the RAID 5 volume, you need to initialize and format it:

a. When prompted, initialize the disks using the default partition style (usually GPT).

b. Format the RAID 5 volume with your desired file system (e.g., NTFS) and assign a drive letter or mount point.

Output:





Result:

Thus, The RAID 5 Volume has been Created and Run Successfully

Ex. No:3(a)	Desktop Virtualization using VNC
DATE:	

Aim:

To Create Desktop Virtualization using VNC

Procedure:

Step 1: Install VNC Server.

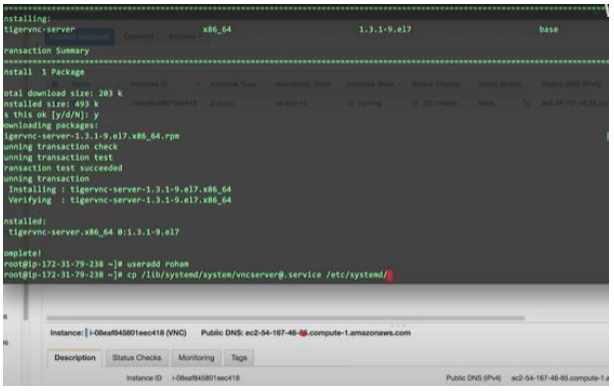
Step 2: Configure VNC Server: After installing the VNC server, you'll need to configure it by setting a password or access control options to secure the remote access. Ensure that the VNC server is running and ready to accept incoming connections.

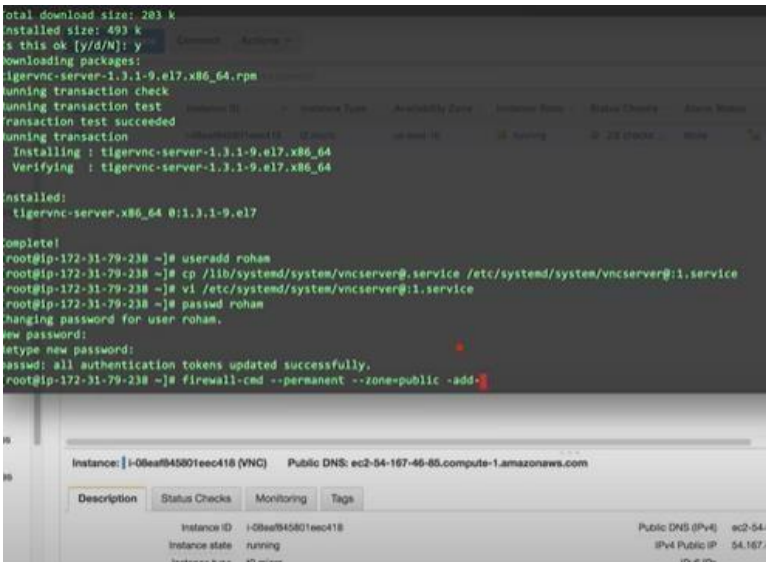
Step 3: Install VNC Viewer.

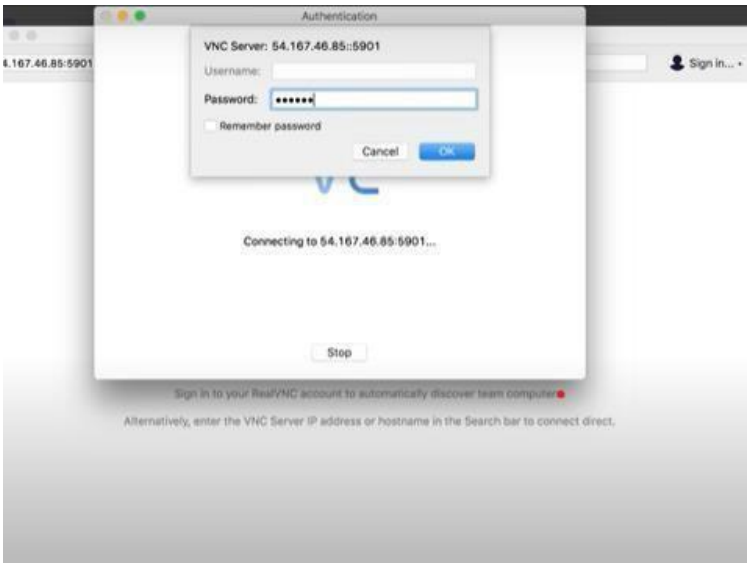
Step 4: Connect to the Remote Desktop: Launch the VNC viewer and enter the IP address or host name of the remote computer (the host) you want to connect to. If you have configured a password or access control options on the VNC server, you will be prompted to enter the credentials. Once authenticated, the VNC viewer establishes a connection to the remote desktop.

Step 5: Control the Remote Desktop: After the connection is established, you will see the remote desktop environment displayed in the VNC viewer. You can now interact with the remote desktop as if you were physically sitting in front of it. You can run applications, access files, and perform any tasks on the remote computer just as if you were using it directly.

Output:







Result:

Thus, The Desktop Virtualization using VNC has been Created and Ru Successfully.

Ex. No:3(b)	Desktop Virtualization using Chrome Remote Desktop
DATE:	

Aim:

To Create Desktop Virtualization using Chrome Remote Desktop

Procedure:

Step 1: Install Chrome Remote Desktop Extension: Make sure you have the Google Chrome web browser installed on both the computer you want to access remotely (the host) and the computer or device you want to use for remote access (the client). Install the "Chrome Remote Desktop" extension from the Chrome Web Store on both devices.

Step 2: Set Up Host Computer (Computer to be Accessed):

a. Open Google Chrome on the host computer.

b. In the address bar, type "chrome://apps" and press Enter.

c. Click on the "Chrome Remote Desktop" icon to open the application.

d. Follow the on-screen instructions to grant necessary permissions and set up remote access for the host computer.

e. Create a secure PIN to use for remote access authentication.

Step 3: Access Host Computer (Client Device):

a. Open Google Chrome on the client device (the computer or device from which you want to access the host computer remotely).

b. In the address bar, type "remotedesktop.google.com/access" and press Enter.

c. Click on the "Access" button under the "Remote Access" section.

d. Sign in with your Google Account (the same account used on the host computer).

Step 4: Choose the Host Computer:

a. After signing in, you should see a list of available computers set up for remote access. Choose the host computer you want to access.

Step 5: Authenticate and Connect:

a. If you have set up a PIN for the host computer, you will be prompted to enter it to authenticate the remote access.

b. Once authenticated, the remote connection will be established, and you will see the host computer's desktop in the Chrome browser window.

Step 6: Control the Host Computer:

a. You can now interact with the host computer's desktop through the Chrome browser on your client device. Use your mouse and keyboard to control the remote desktop.

b. To switch between full-screen and windowed mode, click the "Toggle full screen" button on the top right corner of the remote desktop window.

Step 7: End the Remote Session: To end the remote session, click the "Stop Sharing" button at the bottom of the remote desktop window.

Output:

Result:

Thus, The Desktop Virtualization using Chrome Remote Desktop was Created Successfully

Ex. No:4	Create type 2 virtualization on ESXI 6.5 server
DATE:	

Aim:

To Create type 2 virtualization on ESXI 6.5 server

Procedure:

Step 1: Install ESXi 6.5: First, you need to install VMware ESXi 6.5 on your server.

Step 2: Access ESXi Web Client: Once ESXi is installed and running, access the ESXi Web Client through a web browser on a separate computer. Enter the IP address or hostname of your ESXi server to log in to the management interface.

Step 3: Create a Virtual Machine: To create a virtual machine within ESXi, follow these steps:

a. In the ESXi Web Client, navigate to the "Hosts and Clusters" view.

b. Select your ESXi server from the list of hosts.

c. Click on the "Create/Register VM" button or right-click on the host and choose "New Virtual Machine."

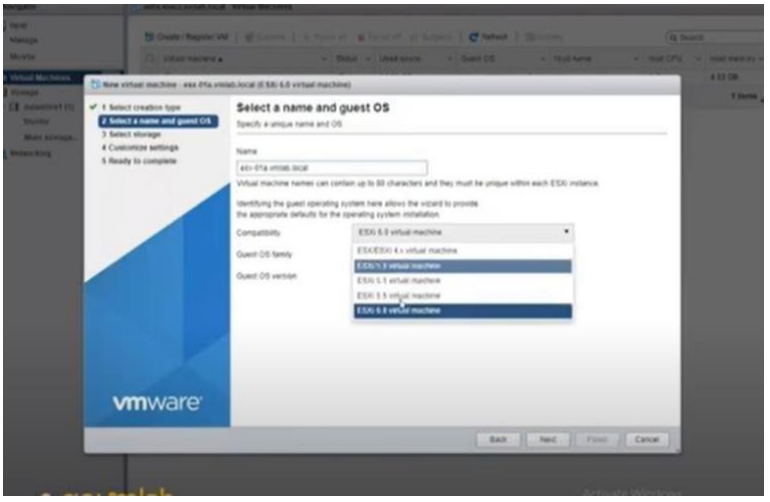
d. The "Create New Virtual Machine" wizard will appear. Follow the steps in the wizard to configure the virtual machine, including providing a name, selecting the guest operating system, setting the desired resources (CPU, memory, disk space, etc.), and selecting a storage location for the virtual machine files.

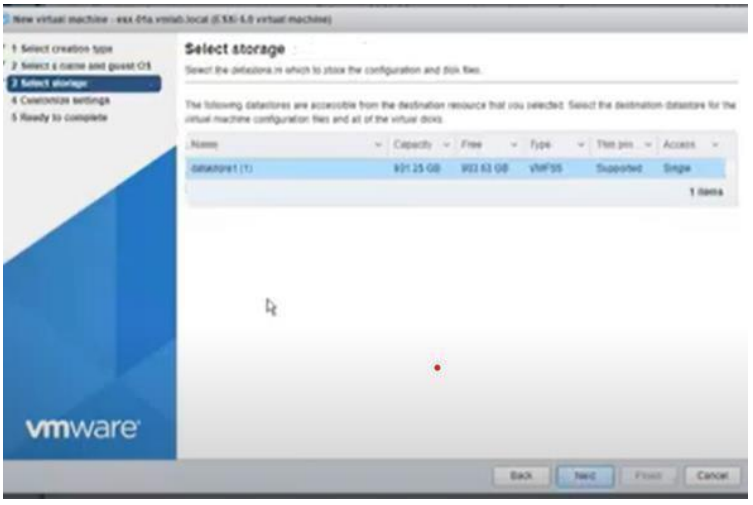
e. Finish the wizard, and the virtual machine will be created.

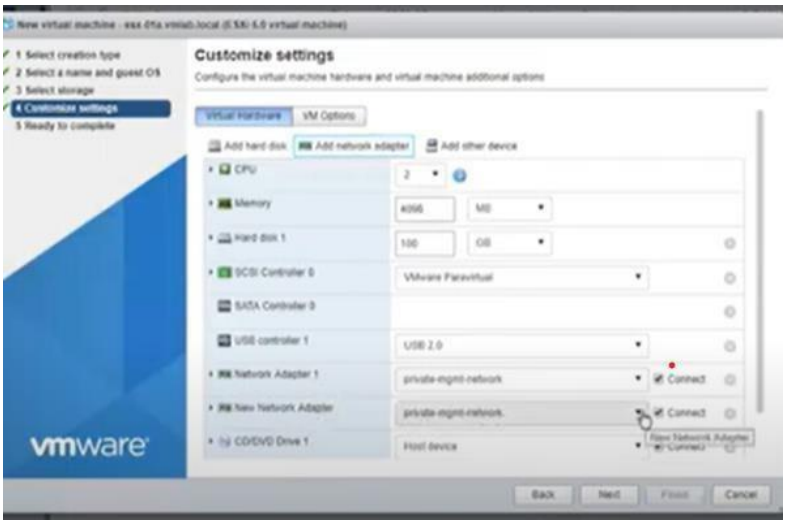
Step 4: Install Guest Operating System: Once the virtual machine is created, you can power it on and install the guest operating system of your choice (e.g., Windows, Linux, etc.) using an ISO image or CD/DVD.

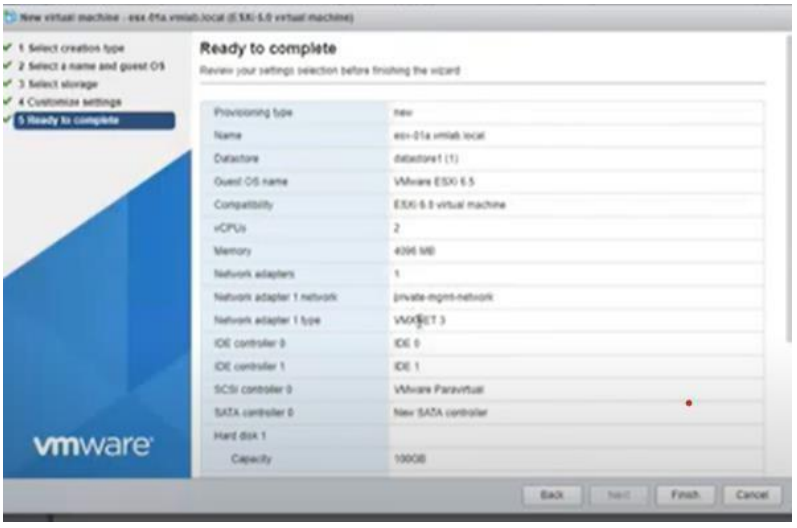
Step 5: Access and Manage the Virtual Machine: After the guest operating system is installed, you can access the virtual machine through the VMware Remote Console (VMRC) or use remote access tools like VNC or RDP to interact with the guest operating system.

OUTPUT:









Result:

Thus, The type 2 virtualization on ESXI 6.5 server was Created and Run Successfully.

Ex. No:5	Create a VLAN in CISCO packet tracer
DATE:	

Aim:

To Create a VLAN in CISCO packet tracer

Procedure:

Step 1: Launch Cisco Packet Tracer: Start Cisco Packet Tracer and open a new or existing project.

Step 2: Add Switches: Drag and drop switches from the "Switches" section of the "Devices" pane onto the workspace. You can use any Cisco switch model available in Packet Tracer.

Step 3: Connect Switches: Connect the switches using Ethernet cables. Click on the Ethernet interfaces of the switches and then click on the other switch's Ethernet interfaces to create connections.

Step 4: Access the CLI: Double-click on the first switch to access its CLI (Command Line Interface). The CLI window will open.

Step 5: Configure VLANs: Enter the configuration mode by typing the following command:

enable

configure terminal

Step 6: Create VLANs: To create VLANs, use the **vlan** command followed by the VLAN number. For example, to create VLAN 10 and VLAN 20, you would do:

vlan 10

vlan 20

Step 7: Assign Ports to VLANs: To assign ports to VLANs, use the

interface command followed by the port number and then the **switchport access vlan** command. For example, to assign port 1 to VLAN 10 and port 2 to VLAN 20, you would do:

interface FastEthernet0/1

switchport mode access

switchport access vlan 10

interface FastEthernet0/2

switchport mode access

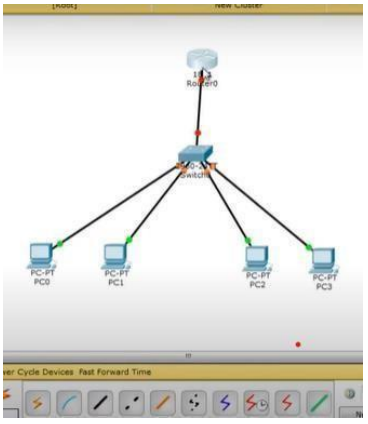
switchport access vlan 20

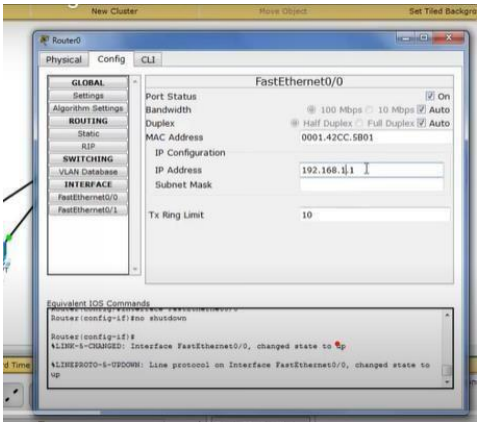
Step 8: Exit Configuration Mode: After assigning VLANs to the desired ports, exit the configuration mode by typing **exit**.

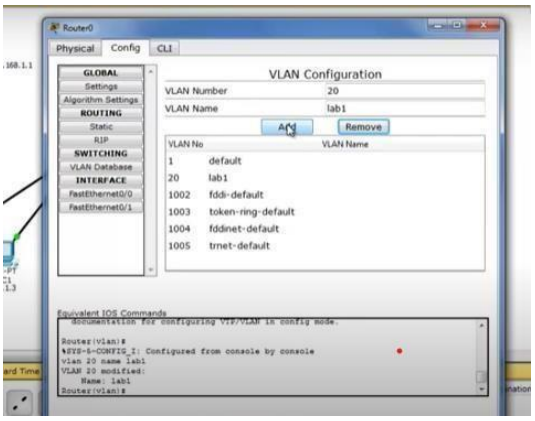
Step 9: Save Configuration: Save the configuration by typing **write** or **copy running-config startup-config** to make sure the changes are saved.

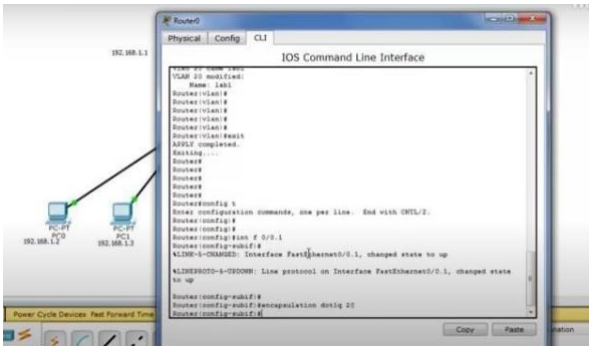
Step 10: Test VLANs: Now that you have created VLANs and assigned ports, you can test the connectivity between devices connected to the switches. Devices in the same VLAN should be able to communicate with each other, while devices in different VLANs should not have direct communication unless you configure inter-VLAN routing.

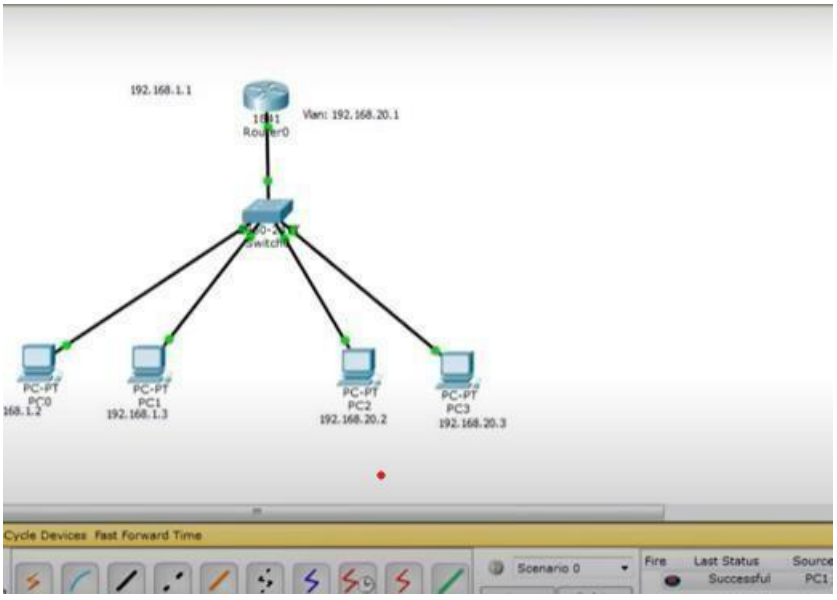
Output:











Result:

Thus, The VLAN in CISCO packet tracer has been Created and Run Successfully.

Ex. No:6	Install KVM in Linux
DATE:	

Aim:

To Install KVM in Linux

Procedure:

Step 1: Check Hardware Support: Before installing KVM, ensure that your system's CPU supports hardware virtualization extensions (Intel VT-x or AMD-V). You can check this by running the following command:

egrep -c '(vmx|svm)' /proc/cpuinfo

If the output is greater than zero (1 or more), it means your CPU supports hardware virtualization.

Step 2: Install KVM Packages: The package names may vary depending on your Linux distribution. Here are the package names for some popular distributions:

For Ubuntu/Debian:

sudo apt update

sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager

For CentOS/RHEL:

sudo yum install qemu-kvm libvirt virt-install bridge-utils

sudo systemctl enable libvirtd

sudo systemctl start libvirtd

For Fedora:

sudo dnf install @virtualization

sudo systemctl enable libvirtd

sudo systemctl start libvirtd

Step 3: Verify Installation: After installing the required packages, check if KVM kernel modules are loaded correctly:

lsmod | grep kvm

The output should show **kvm** and **kvm_intel** or **kvm_amd** modules loaded, depending on your CPU.

Step 4: Configure Permissions: For non-root users to manage virtual machines using KVM, add them to the **libvirt** group:

sudo usermod -aG libvirt <username>

Remember to log out and log back in for the changes to take effect.

Step 5: Enable Nested Virtualization (Optional): If you plan to run virtual machines with nested virtualization (e.g., running KVM inside a KVM guest), you may need to enable nested virtualization on the host. This step is only required if you intend to run virtual machines with KVM as guests.

For Intel CPUs:

echo "options kvm-intel nested=1" | sudo tee /etc/modprobe.d/kvm-intel.conf

sudo modprobe -r kvm_intel

sudo modprobe kvm_intel

For AMD CPUs:

echo "options kvm-amd nested=1" | sudo tee /etc/modprobe.d/kvm-amd.conf

sudo modprobe -r kvm_amd

sudo modprobe kvm_amd

Step 6: Install and Configure Virt-Manager (Optional): Virt-Manager is a graphical user interface tool to manage virtual machines using KVM. If you prefer a GUI interface, you can install Virt-Manager:

For Ubuntu/Debian:

sudo apt install virt-manager

For CentOS/RHEL:

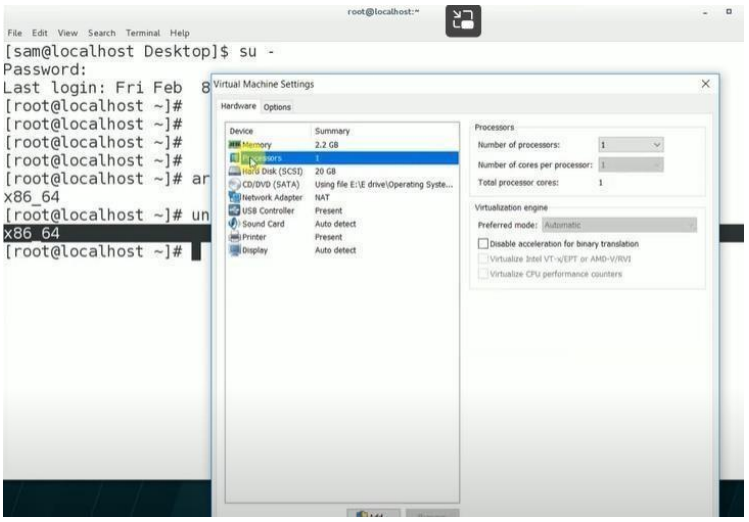
sudo yum install virt-manager

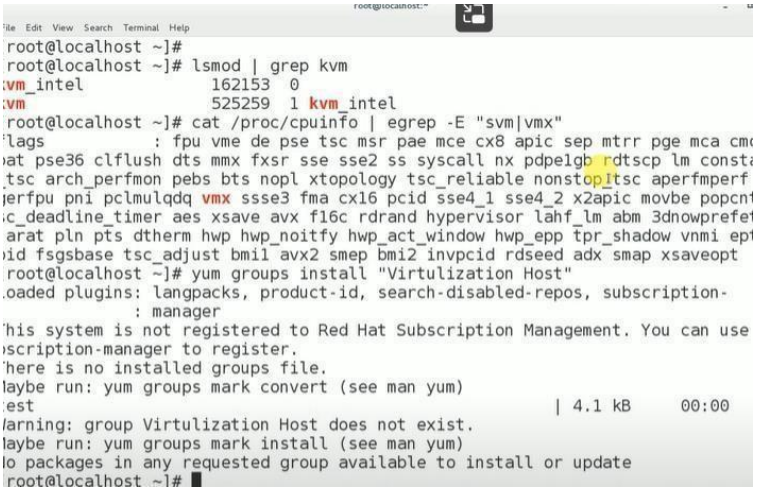
For Fedora:

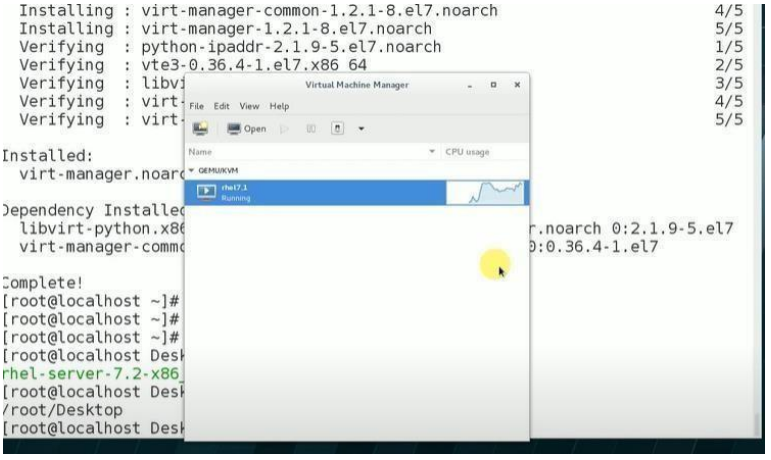
sudo dnf install virt-manager

Once the installation is complete, you can start creating and managing virtual machines using KVM. You can use command-line tools like **virsh** or a graphical interface like Virt-Manager to interact with KVM.

Output:







Result:

Thus, The KVM has been Installed in Linux Successfully.

Ex. No:7	Create Nested Virtual Machine (VM under another VM)
DATE:	

Aim:

To Create Nested Virtual Machine (VM under another VM)

Procedure:

Step 1: Enable Nested Virtualization: Before creating a nested VM, ensure nested virtualization is enabled on the host VM (the VM that will run other VMs).

For VirtualBox:

Open VirtualBox and select the host VM from the list. Go to "Settings" > "System" > "Processor" and check the "Enable Nested VT-x/AMD-V" option.

Step 2: Install an Operating System in the Host VM: Ensure you have an operating system installed in the host VM. This will be the environment in which you'll run the nested VMs.

Step 3: Install VirtualBox within the Host VM: Inside the host VM, download and install VirtualBox (or any other virtualization software you prefer) as if you were installing it on a physical machine.

Step 4: Create the Nested VM: Now that you have VirtualBox installed within the host VM, you can create a new VM inside it.

a. Open VirtualBox within the host VM.

b. Click on "New" to create a new VM.

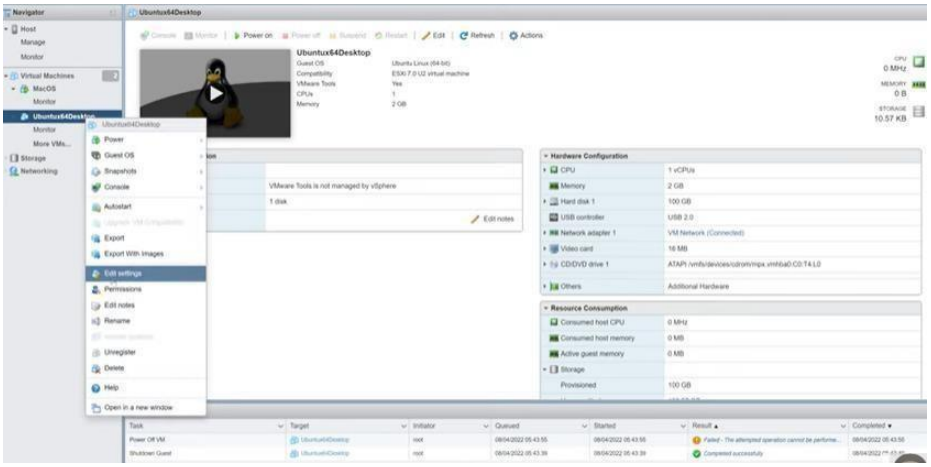
c. Follow the VM creation wizard to set up the nested VM, including selecting the guest operating system, allocating memory, creating a virtual hard disk, etc.

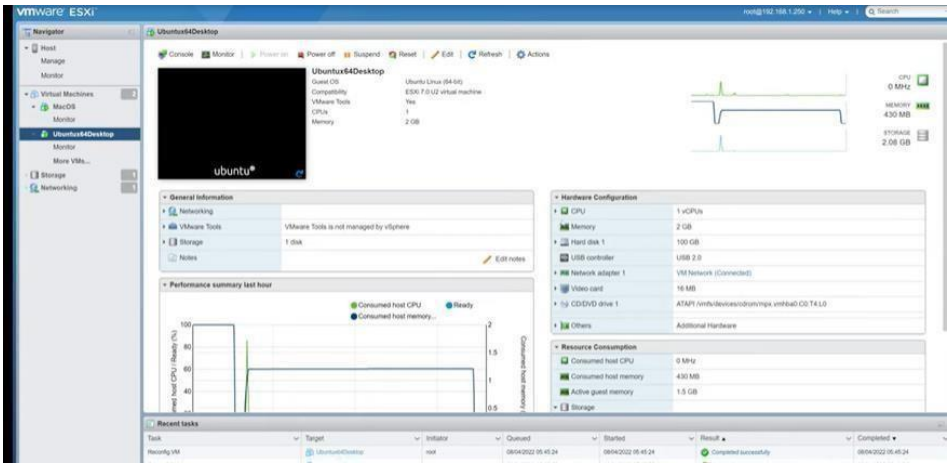
Step 4: Install the Guest Operating System in the Nested VM: With the nested VM created, start it and install the guest operating system, just as you would with any regular VM installation.

Step 5; Configure Networking (Optional): Depending on your requirements, you may need to configure the networking of the nested VM to allow communication with other VMs or external networks.

Step 6: Use the Nested VM: Once the nested VM is set up and the guest operating system is installed, you can use it just like any other VM. Install applications, run tests, or perform any tasks within the nested VM.

Output:





Result:

Thus, The Nested Virtual Machine (VM under another VM) was Created and Run Successfully.

Ex. No:8	Install a c compiler in the virtual machine
DATE:	

AIM:

To Install a C compiler in the virtual machine and execute a sample C program.

PROCEDURE :

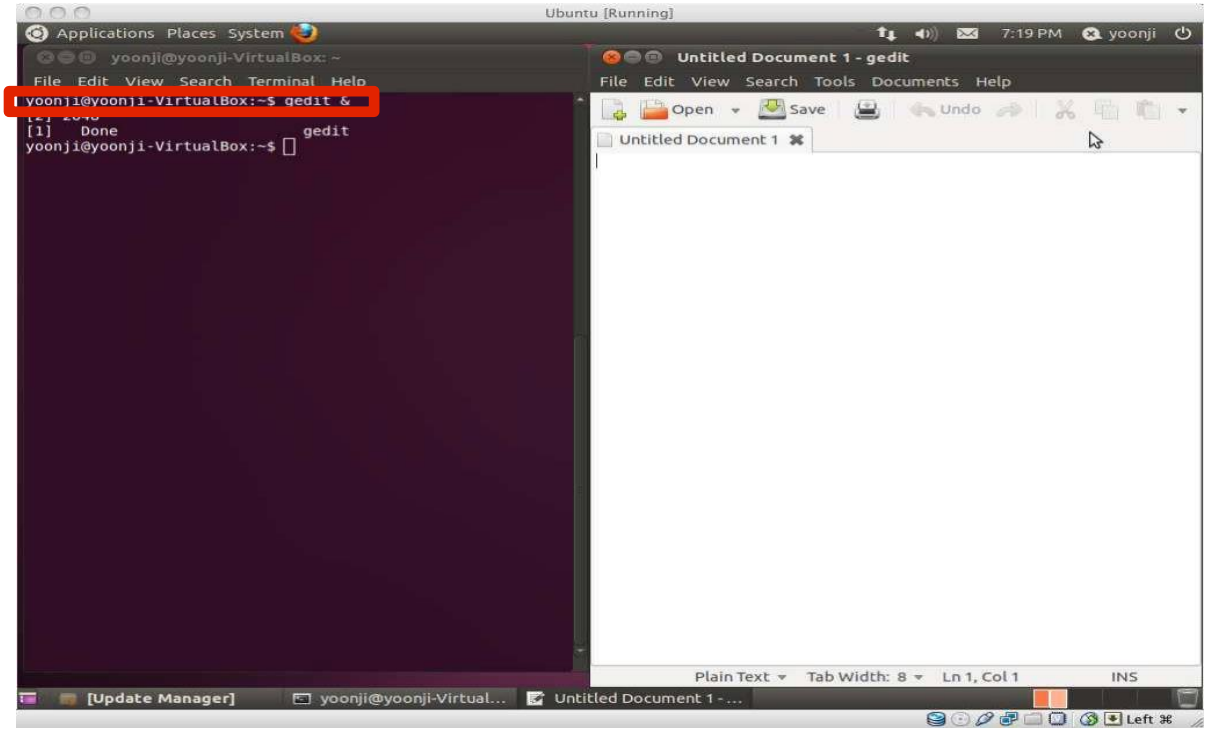
Step 1 : Open Terminal (Applications-Accessories-Terminal)

Step 2 : Open **gedit** by typing “gedit &” on terminal

(You can also use any other Text Editor application)

Step 3: Type the following on gedit (or any other text editor)

```
#include<3tdio.h> main()
{
printf("Hello World\n");
}
```

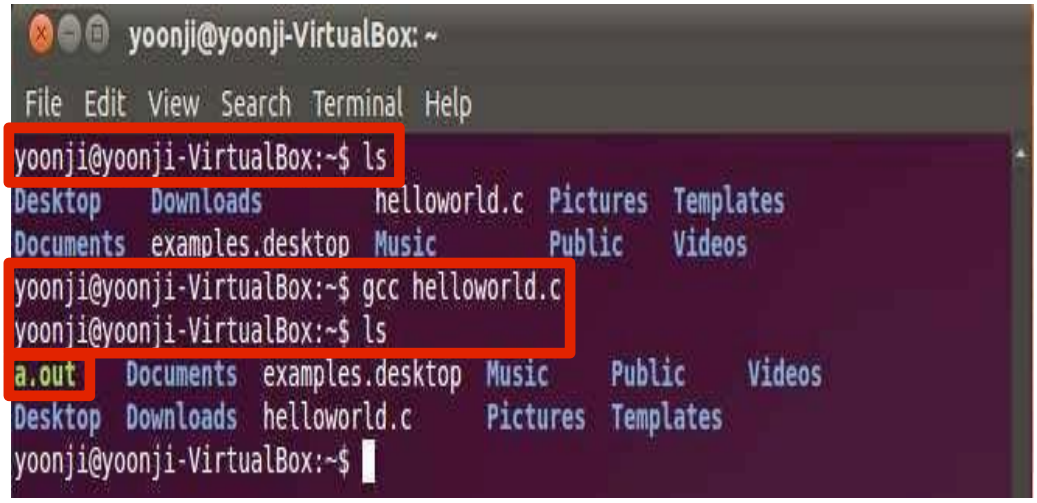


Save this file as “helloworld.c”

Step 4 : Type “ls” on Terminal to see all files under current folder

Confirm that “helloworld.c” is in the current directory. If not, type cd DIRECTORY_PATH to go to the directory that has “helloworld.c”

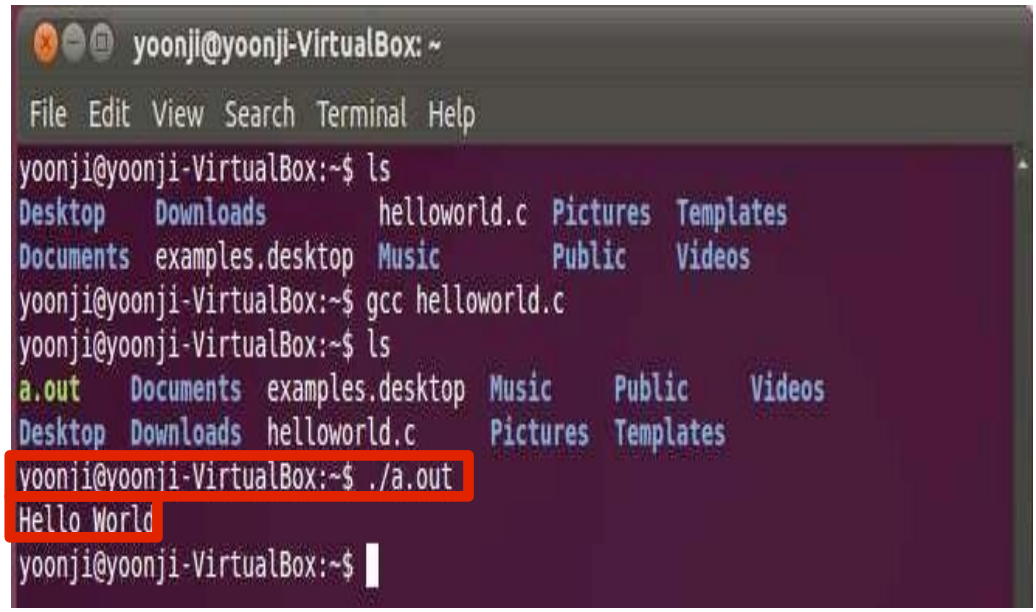
Step 5 :Type “gcc helloworld.c” to compile, and type “ls” to confirm that a new executable



file “a.out” is created.

Step 6 : Type “./a.out” on Terminal to run the program

OUTPUT :



Result:

Thus, **C** compiler is installed in the virtual machine and executed a sample program Successfully.

Ex. No:9	Install Google App Engine
DATE:	

AIM: To create hello world app and other simple web applications using python/java.

PROCEDURE :

STEP 1 : PRE--REQUISITES: PYTHON 2.5.4
The App Engine SDK allows you to run Google App Engine Applications on your local computer. It simulates the run---time environment of the Google App Engine infrastructure.

If you don't already have Python 2.5.4 installed in your computer, download andInstall Python 2.5.4 from:

http://www.python.org/download/releases/2.5.4/

STEP 2 : DOWNLOAD AND INSTALL

You can download the Google App Engine SDK by going to:

http://code.google.com/appengine/downloads.html

and download the appropriate install package.

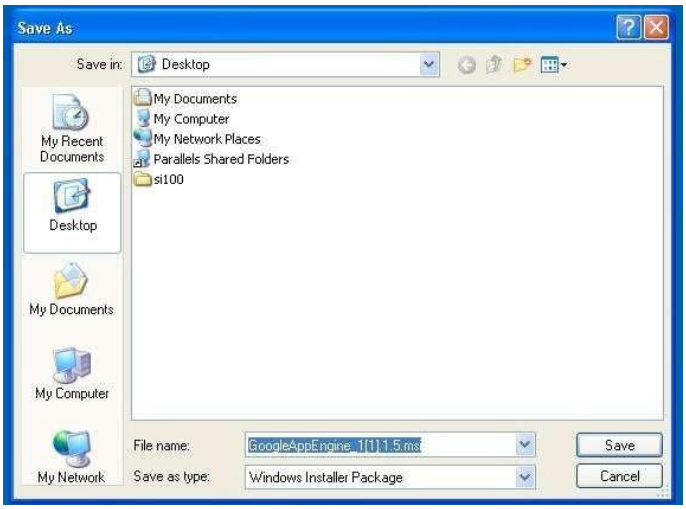
Download the Google App Engine SDK

Before downloading, please read the Terms that govern your use of the App Engine SDK.

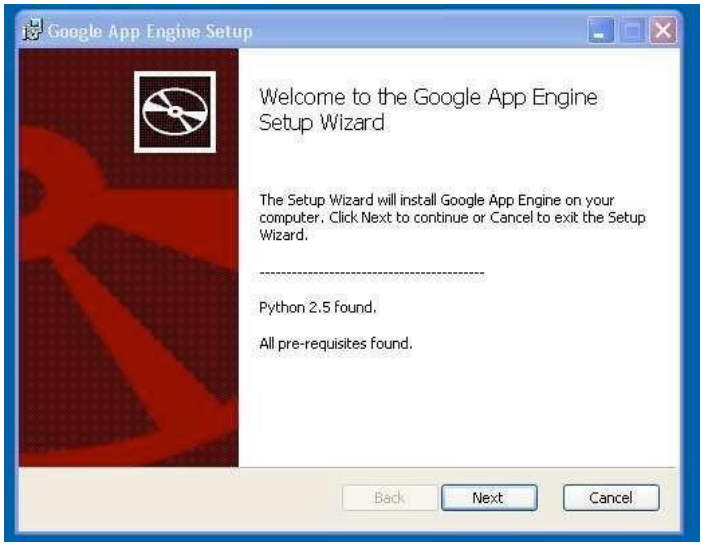
Please note: The App Engine SDK is under **active development**, please keep this in mind as you explore its capabilities. See the SDK Release Notes for the information on the most recent changes to the App Engine SDK. If you discover any issues, please feel free to notify us via our Issue Tracker.

Platform	Version	Package	Size	SHA1 Checksum
Windows	1.1.5 - 10/03/08	GoogleAppEngine_1.1.5.msi	2.5 MB	e974312b4aefc0b3873ff0d93eb4c525d5e88c30
Mac OS X	1.1.5 - 10/03/08	GoogleAppEngineLauncher-1.1.5.dmg	3.6 MB	f62208ac01c1b3e39796e58100d5f1b2f052d3e7
Linux/Other Platforms	1.1.5 - 10/03/08	google_appengine_1.1.5.zip	2.6 MB	cbb9ce817bdabf1c4f181d9544864e55ee253de1

Download the Windows installer – the simplest thing is to download it to yourDesktop or another folder that you remember.



Double Click on the **GoogleApplicationEngine** installer.



Click through the installation wizard, and it should install the App Engine. If you donot have Python 2.5, it will install Python 2.5 as well.

Once the install is complete you can discard the downloaded installer

STEP 3 : MAKING YOUR FIRST APPLICATION

Now you need to create a simple application. We could use the "+" option to have the launcher make us an application – but instead we will do it by hand to get a better sense of what is going on.

Make a folder for your Google App Engine applications. Make the Folder on Desktop called "**apps**" – the path to this folder is:

C:\Documents and Settings\csev\Desktop\apps

And then make a sub---folder in within **apps** called "**ae--01--trivial**" – the path to this folder would be:

C:\ Documents and Settings \csev\Desktop\apps\ae--01--trivial

Using a text editor such as JEdit (www.jedit.org), create a file called **app.yaml** in the **ae--01--trivial** folder with the following contents:

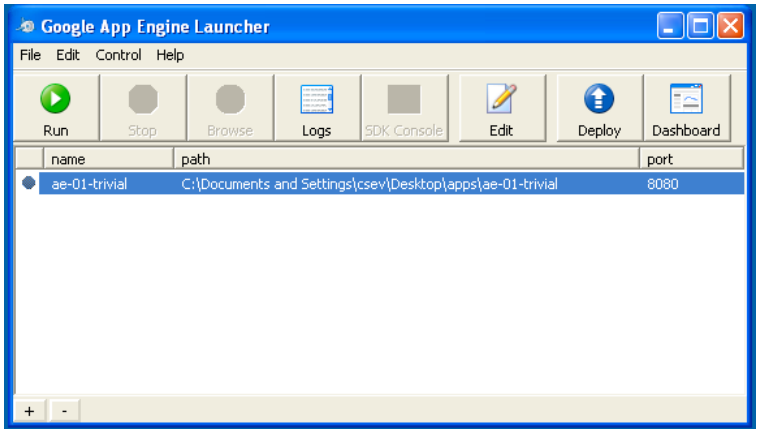
```
application: ae-01-trivialversion: 1
runtime: pythonapi_version: 1

handlers:
- url: /.*
    script: index.py
```

Note: Please do not copy and paste these lines into your text editor – you might endup with strange characters – simply type them into your editor.

Then create a file in the **ae--01--trivial** folder called **index.py** with three lines in it:

```
print 'Content-Type: text/plain'print ' '
print 'Hello there Chuck'
```

Then start the **GoogleAppEngineLauncher** program that can be found under **Applications**. Use the **File --> Add Existing Application** command and navigate into the **apps** directory and select the **ae--01--trivial** folder. Once you have addedthe application, select it so that you can control the application using the launcher.

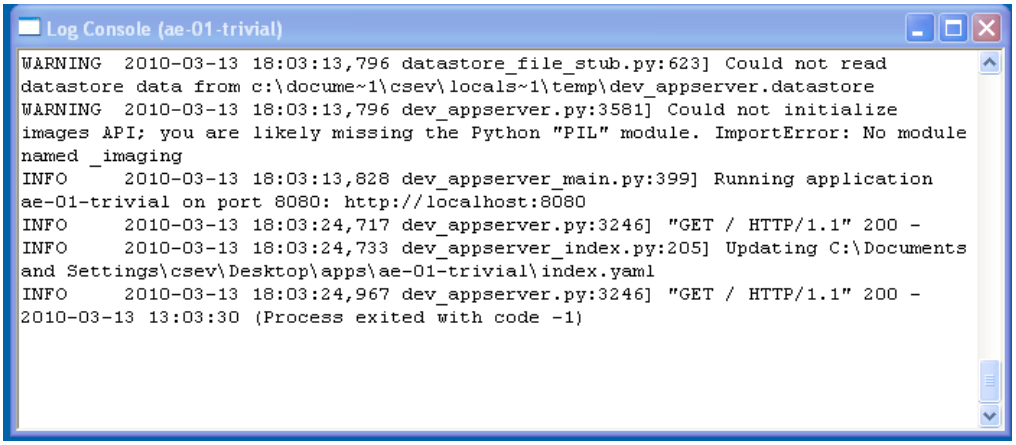


Once you have selected your application and press **Run**. After a few moments your application will start and the launcher will show a little green icon next to your application. Then press **Browse** to open a browser pointing at your application which is running at **http://localhost:8080/**

STEP 4 : Paste **http://localhost:8080** into your browser and you should see yourapplication as follows: Just for fun, edit the **index.py** to change the name "Chuck" to your own name andpress Refresh in the browser to verify your updates.

Watching the Log

You can watch the internal log of the actions that the web server is performing whenyou are interacting with your application in the browser. Select your application in the Launcher and press the **Logs** button to bring up a log window:



Each time you press **Refresh** in your browser – you can see it retrieving the outputwith a **GET** request.

Shutting Down the Server

To shut down the server, use the Launcher, select your application and press the **Stop** button.

OUTPUT :

RESULT :

Thus, Google App Engine is installed and a hello world app was run using python.

Ex. No:10	Files transfer from one virtual machine to another virtual machine
DATE:	

AIM : To find a procedure to transfer the files from one virtual machine to another virtual machine.

PROCEDURE :

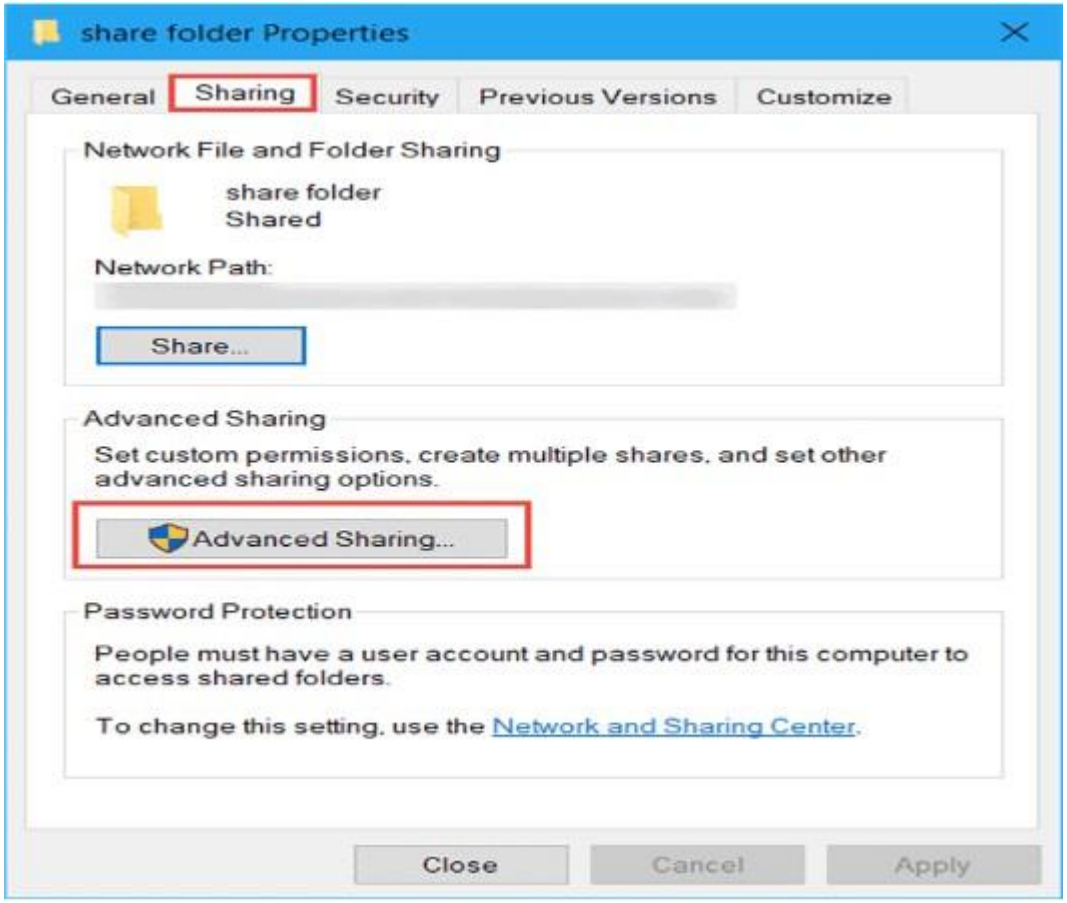
1. You can copy few (or more) lines with ***copy & paste*** mechanism.
   For this you need to share clipboard between host OS and guest OS, installing **Guest Addition** on both the virtual machines (probably setting *bidirectional* and restarting them).You *copy* from *guest OS* in the clipboard that is shared with the *host OS*.
   Then you *paste* from the *host OS* to the second *guest OS.*
2. You can enable **drag and drop** too with the same method (Click on the machine, settings,general, advanced, drag and drop: set to *bidirectional*)
3. You can have **common *Shared Folders*** on both virtual machines and use one of thedirectory shared as buffer to copy.
   Installing **Guest Additions** you have the possibility to set Shared Folders too. As you put afile in a shared folder from *host OS* or from *guest OS*, is immediately visible to the other. (Keep in mind that can arise some problems for date/time of the files when there are different clock settings on the different virtual machines).
   If you use the same folder shared on more machines you can exchange files directly copyingthem in this folder.
4. You can use **usual method to copy files between 2 different computer** with client-server application. (e.g. scp with sshd active for linux, winscp... you can get some info about SSHservers e.g. here)
   You need an active server (sshd) on the receiving machine and a client on the sending machine. Of course you need to have the authorization setted (via password or, better, viaan automatic authentication method).
   Note: many Linux/Ubuntu distribution install sshd by default: you can see if it is runningwith pgrep sshd from a shell. You can install with sudo apt-get install openssh-server.
5. You can **mount part of the file system** of a virtual machine via NFS or SSHFS on theother, or you can **share file and directory** with Samba.
   You may find interesting the article Sharing files between guest and host withoutVirtualBox shared folders with detailed step by step instructions.

You should remember that you are dialling with a little network of machines with differentoperative systems, and in particular:

- Each virtual machine has its own operative system running on and acts as a physicalmachine.
- Each virtual machine is an instance of a program *owned* by an *user* in the hosting operativesystem and should undergo the restrictions of the *user* in the *hosting OS*.
  E.g Let we say that Hastur and Meow are users of the hosting machine, but they did not allow each other to see their directories (no read/write/execute authorization). When each ofthem run a virtual machine, for the hosting OS those virtual machine are two normal programs owned by Hastur and Meow and cannot see the private directory of the other user.This is a restriction due to the *hosting OS*. It's easy to overcame it: it's enough to give authorization to read/write/execute to a directory or to chose a different directory in which both users can read/write/execute.

- Windows likes mouse and Linux fingers. :-)
  I mean I suggest you to enable *Drag & drop* to be cosy with the Windows machines andthe *Shared folders* or to be cosy with Linux.
  When you will need to be fast with Linux **you will feel the need** of ssh-keygen and
  to Generate once SSH Keys to copy files on/from a remote machine without writingpassword anymore. In this way it functions bash auto-completion remotely too!

OUTPUT



RESULT :

Thus, the files are transferred from one virtual machine to another virtual machine successfully**.**